

RMDL Assessment 2: Data Analysis Report

YOUR EXAM NUMBER

0.1 Background

- *Instructions are given in italics. You should remove all instructions before submitting.*
- *You MUST include all code and code output in this report. This is a data analysis report, not an essay or journal article.*

Write a brief literature background for your Stroop task study. Explicitly state your Research Questions and, if available, your Research Hypotheses.

0.1.1 Research Questions/Hypotheses

0.2 Methods

Briefly report the methods of the study (participants, materials, procedure, statistical analysis...). Note that when discussing the statistical analysis you should clearly state how the analysis results will be used to assess the RQs/RHs (be as specific and precise as you can). Include an Open Research/Data availability statement.

0.3 Data

Remember to include all code and code output in the submission.

0.3.1 Read and process the data

Read the data here. If the data needs processing, like mutating columns or creating new ones, briefly explain why processing is needed and what you did. Include a description of each of the relevant columns in the data.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(brms)
```

Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing `help('brms')`. A more detailed introduction
to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
library(dplyr)
library(posterior)
```

This is posterior version 1.6.1

Attaching package: 'posterior'

The following objects are masked from 'package:stats':

mad, sd, var

The following objects are masked from 'package:base':

%in%, match

```
library(tidybayes)
```

Warning: package 'tidybayes' was built under R version 4.5.3

Attaching package: 'tidybayes'

The following objects are masked from 'package:brms':

dstudent_t, pstudent_t, qstudent_t, rstudent_t

```
alldata = list.files(  
  path = "./data/",  
  pattern = "*.csv$",  
  recursive = TRUE ,  
  full.names = TRUE  
)  
  
compiledlist <- lapply(alldata, function(x) {  
  read_csv(x, skip = 3)  
})
```

Rows: 40 Columns: 26

-- Column specification -----

Delimiter: ","

chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...

dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

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```

```

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i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```

finaldata <- bind_rows(compiledlist) |>
  filter(condition1 != "practise") |>
  mutate(
    early_late = case_when(
      substring(id,1,1) == "1" ~ "early",
      substring(id,1,1) == "2" ~ "late",
      TRUE ~ ""
    )
  )

```

```

newfinaldata <- finaldata %>%
  separate_wider_delim(
    condition1,
    delim = "_",
  )

```

```
names = c("Condition", "Language")
)
```

```
newfinaldata$Language <- recode(newfinaldata$Language,
                                "eng" = "English",
                                "fr"  = "French")

newfinaldata$early_late <- recode(newfinaldata$early_late,
                                  "early" = "Early bilingual",
                                  "late"  = "Late bilingual")

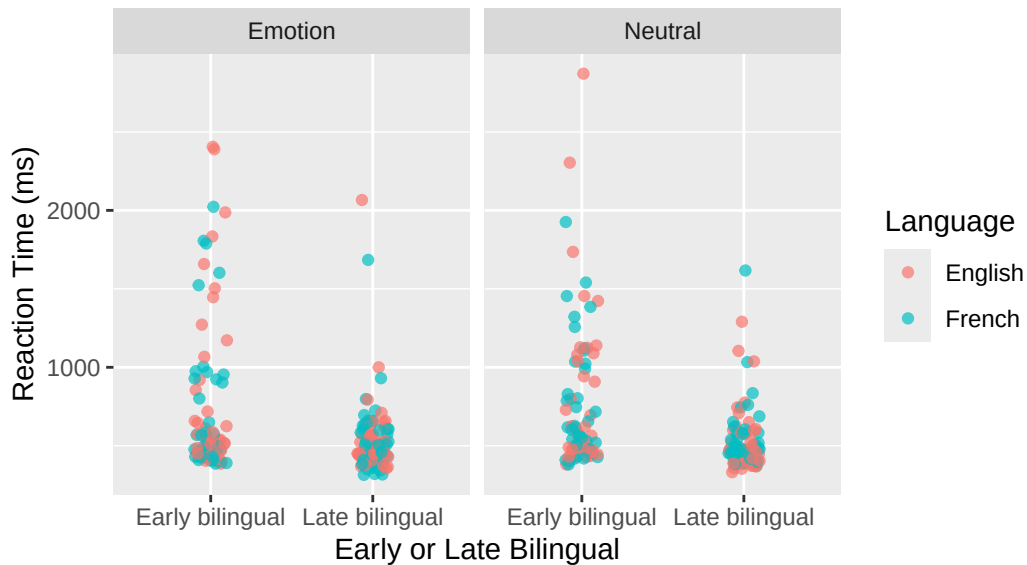
newfinaldata$Condition <- recode(newfinaldata$Condition,
                                  "emotion" = "Emotion",
                                  "neutral"  = "Neutral")
```

0.3.2 Plot the data

Create relevant plots: for each, include a description of the plot (type, axes, colours used and so on) and a description of relevant patterns.

```
newfinaldata |>
  ggplot(aes(early_late, RT, colour = Language)) +
  geom_jitter(alpha = 0.7, width = 0.1) +
  facet_wrap(~ Condition) +
  labs(
    x = "Early or Late Bilingual",
    y = "Reaction Time (ms)",
    title = "AoA (Early or Late Bilingual) influence on reaction time
            to emotion and to neutral words")
```

AoA (Early or Late Bilingual) influence on reaction time to emotion and to neutral words



0.3.3 Summarise the data

Calculate and report appropriate summary measures for relevant variables in the data.

```
newfinaldata |>
  group_by(Condition, Language, early_late) |>
  summarise(
    mean = mean(RT), sd = sd(RT))
```

`summarise()` has grouped output by 'Condition', 'Language'. You can override using the `.groups` argument.

```
# A tibble: 8 x 5
# Groups:   Condition, Language [4]
  Condition Language early_late    mean    sd
  <chr>      <chr>    <chr>    <dbl> <dbl>
1 Emotion   English  Early bilingual  818.  561.
2 Emotion   English  Late bilingual   542.  255.
3 Emotion   French   Early bilingual  726.  440.
4 Emotion   French   Late bilingual   533.  216.
5 Neutral   English  Early bilingual  806.  537.
```

6	Neutral	English	Late bilingual	516.	196.
7	Neutral	French	Early bilingual	756.	377.
8	Neutral	French	Late bilingual	537.	204.

0.4 Regression modelling

Write an explanation of the regression model(s) you will run to answer the chosen research question. Fit the model and get a model summary and relevant model plots. Report the results of the model.

```
brm_newfinaldata <- brm(
  RT ~ early_late * Language * Condition +
    (Language + Condition | id ),
  family = lognormal,
  data = newfinaldata,
  seed = 456,
  cores = 4,
  file = "cache/brmfinaldata",
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)
```

```
summary(brm_newfinaldata, prob = 0.8)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

```
Multilevel Hyperparameters:
~id (Number of levels: 11)
```

	Estimate	Est.Error	1-80% CI	u-80% CI	Rhat
sd(Intercept)	0.40	0.12	0.28	0.55	1.00
sd(LanguageFrench)	0.05	0.04	0.01	0.11	1.00
sd(ConditionNeutral)	0.06	0.05	0.01	0.13	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.85	0.37	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.71	0.53	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.71	0.63	1.00
Bulk_ESS Tail_ESS					
sd(Intercept)	1410	1785			

sd(LanguageFrench)	1939	1869
sd(ConditionNeutral)	1508	1335
cor(Intercept,LanguageFrench)	3895	2755
cor(Intercept,ConditionNeutral)	4083	2955
cor(LanguageFrench,ConditionNeutral)	2487	2824

Regression Coefficients:

	Estimate	Est.Error			
Intercept	6.54	0.18			
early_lateLatebilingual	-0.29	0.25			
LanguageFrench	-0.08	0.07			
ConditionNeutral	0.01	0.07			
early_lateLatebilingual:LanguageFrench	0.07	0.09			
early_lateLatebilingual:ConditionNeutral	-0.05	0.09			
LanguageFrench:ConditionNeutral	0.07	0.08			
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12			
	1-80% CI	u-80% CI	Rhat		
Intercept	6.31	6.76	1.00		
early_lateLatebilingual	-0.59	0.01	1.01		
LanguageFrench	-0.16	0.00	1.00		
ConditionNeutral	-0.08	0.09	1.00		
early_lateLatebilingual:LanguageFrench	-0.05	0.18	1.00		
early_lateLatebilingual:ConditionNeutral	-0.17	0.07	1.00		
LanguageFrench:ConditionNeutral	-0.04	0.17	1.00		
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.15	0.14	1.00		
	Bulk_ESS	Tail_ESS			
Intercept	965	1898			
early_lateLatebilingual	981	1825			
LanguageFrench	2101	2546			
ConditionNeutral	2164	2434			
early_lateLatebilingual:LanguageFrench	2167	2787			
early_lateLatebilingual:ConditionNeutral	2229	2620			
LanguageFrench:ConditionNeutral	2156	2769			
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687			

Further Distributional Parameters:

	Estimate	Est.Error	1-80% CI	u-80% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.26	0.29	1.00	5596	3054

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```

draws_brm_newfinaldata <- as_draws_df(brm_newfinaldata)

frame_draws_brm_newfinaldata <- draws_brm_newfinaldata |>
  mutate(
    eng_emotion_early = b_Intercept,
    fr_emotion_early = b_Intercept + b_LanguageFrench,
    eng_emotion_late = b_Intercept + b_early_lateLatebilingual,
    eng_neutral_early = b_Intercept + b_ConditionNeutral,
    fr_emotion_late = b_Intercept + b_early_lateLatebilingual + b_LanguageFrench + `b_early_...
    eng_neutral_late = b_Intercept + b_ConditionNeutral + b_early_lateLatebilingual + `b_ear...
    fr_neutral_early = b_Intercept + b_LanguageFrench + b_ConditionNeutral + `b_LanguageFren...
    fr_neutral_late = b_Intercept + b_ConditionNeutral + b_LanguageFrench + b_early_lateLate...
  ) |>
  select(eng_emotion_early, fr_emotion_early, eng_emotion_late, eng_neutral_early, fr_emotion...

```

Warning: Dropping 'draws_df' class as required metadata was removed.

```

plot2 <- frame_draws_brm_newfinaldata |>
  select(eng_emotion_early, fr_emotion_early, eng_emotion_late, eng_neutral_early, fr_emotion...
  pivot_longer(
    eng_emotion_early:fr_neutral_late,
    names_to = "cond",
    values_to = "epred"
  ) |>
  separate(cond, c("Language", "Condition", "Early_Late"))

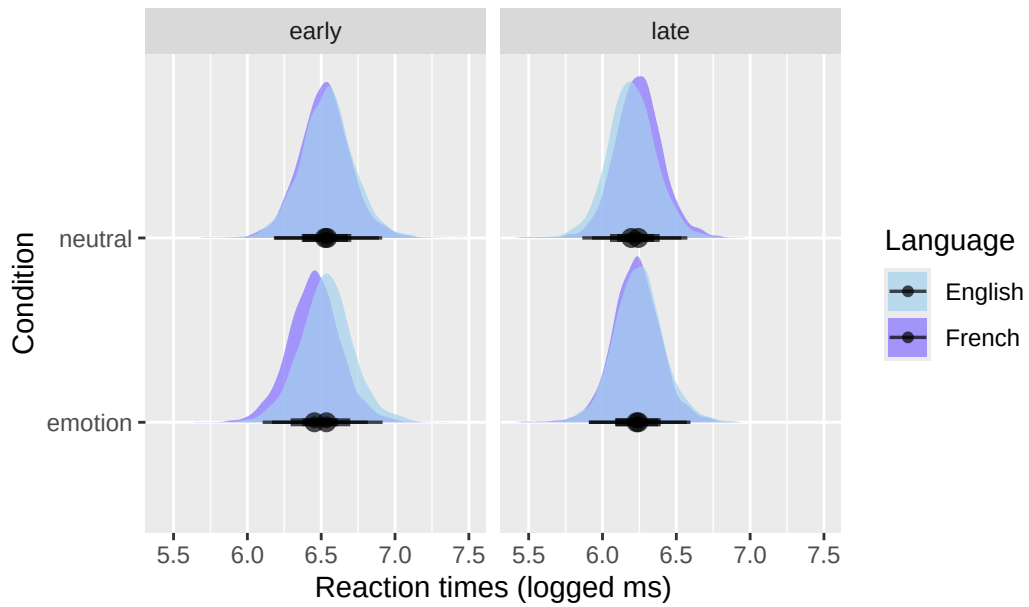
```

```

plot2 |>
  ggplot(aes(epred, Condition, fill = Language )) +
  stat_halfeye(alpha = 0.7) +
  scale_fill_manual(
    labels = c("English", "French"),
    values = c("lightskyblue2", "slateblue1")
  ) + facet_grid(cols = vars(Early_Late)) +
  labs(title = "Reaction times (logged) across conditions for early and late bilinguals", x = ...

```

Reaction times (logged) across conditions for early and late



0.5 Discussion

Write a brief discussion of the your model results in light of the research question.